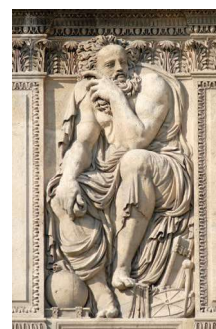


## 4.2 Circumference of a circle

Since the ancient times, people have known about a special number that links a circle's diameter to its circumference. We know this number as pi ( $\pi$ ). Pi is a mathematical constant that appears in formulas relating to circles, but it is also important in many other areas of mathematics. The actual value of pi has been studied and approximated by ancient and more modern civilisations over thousands of years. The Egyptians knew pi was slightly more than 3 and approximated it to be  $\frac{356}{81} \approx 3.16$ . The Babylonians used  $\frac{25}{8} \approx 3.125$  and the ancient Indians used  $\frac{339}{108} \approx 3.139$ .

It is believed that Archimedes of Syracuse (287–212 BCE) was the first person to use a mathematical technique to evaluate pi. He was able to prove that pi was greater than  $\frac{223}{71}$  and less than  $\frac{22}{7}$ . In 480 AD, the Chinese mathematician Zu Chongzhi showed that pi was close to  $\frac{335}{113} \approx 3.1415929$ , which is accurate to six decimal places.

Before the use of calculators, the fraction  $\frac{22}{7}$  was commonly used as a good and simple approximation to pi. Interestingly, mathematicians have been able to prove that pi is an irrational number, which means that there is no fraction that can be found that is exactly equal to pi. If the exact value of pi was written down as a decimal, the decimal places would continue forever with no repeated pattern.



### Let's start: Discovering pi

Here are the diameters and circumferences for 3 circles correct to two decimal places. Use a calculator to work out the value of Circumference  $\div$  Diameter and put your results in the third column. Add your own circle measurements by measuring the diameter and circumference of circular objects such as a can or a wheel.

| Diameter<br>$d$ (mm) | Circumference<br>$C$ (mm) | $C \div d$ |
|----------------------|---------------------------|------------|
| 4.46                 | 14.01                     |            |
| 11.88                | 37.32                     |            |
| 40.99                | 128.76                    |            |
| Add your own         | Add your own              |            |

- What do you notice about the numbers  $C \div d$  in the third column?
- Why might the numbers in the third column vary slightly from one set of measurements to another?
- What rule can you write down which links  $C$  with  $d$ ?

#### ■ Features of a circle

- **Diameter** ( $d$ ) is the distance across the centre of a circle.
- **Radius** ( $r$ ) is the distance from the centre to the circle. Note  $d = 2r$ .

#### ■ Circumference ( $C$ ) is the distance around a circle.

- $C = 2\pi r$  or  $C = \pi d$

#### ■ Pi ( $\pi$ ) $\approx 3.14159$ (correct to five decimal places)

- Common approximations include 3.14 and  $\frac{22}{7}$ .
- A more precise estimate for pi can be found on most calculators or on the internet.

